Segmented Network Lab

Older homelab I built around network segmentation. A Cisco switch and SonicWall handle the network, with ESXi/vCenter on iSCSI storage and Windows Server running AD, DNS and DHCP. The lab has since been torn down. These are my build notes with the reasoning behind each step.

Main goal was segmentation. Users and servers on separate VLANs/zones so if a laptop got compromised it couldn't just reach the servers.

```
Internet
   |
SonicWall X1 (WAN)
   |
SonicWall X0 (trunk: VLAN 1 untagged, VLAN 4 and 5 tagged)
   |
Cisco switch (layer 2 only)
   |-- Fa0/3  access VLAN 4 -> laptop
   |-- Fa0/15 trunk         -> ESXi host -> iSCSI NAS
```

Some screenshots are from the tutorials I followed (linked at the end), so IPs and domain names vary between sections.

Cisco Switch

- Set the native VLAN (VLAN 1) IP in the same range as the SonicWall X0 LAN

  Why: *Management IP so I can SSH into the switch. It sits on the same network as the firewall's LAN port.*

```
interface Vlan1
 ip address 192.168.168.20 255.255.255.0
!
```

- Created the VLAN and added it to the interface for the laptop (Fa0/3, access VLAN 4)

  Why: *The laptop doesn't understand VLAN tags. The access port puts its traffic into VLAN 4.*

```
interface FastEthernet0/3
 switchport access vlan 4
 switchport mode access
!
```

- Admin access: line con / line vty, set privilege and password

  Why: *Locks down console and remote management access to the switch.*

```
line con 0
line vty 0 4
 password
 login local
line vty 5 15
 password
 login local
```

- Enabled SSH: hostname > ip domain-name > crypto key generate rsa

  Why: *SSH is encrypted, telnet isn't. Generating the RSA key needs the hostname and domain name set first.*

Step 2	**configure terminal** **Example:** `Switch# configure terminal`
Step 3	**hostname** *hostname* **Example:** `Switch(config)# hostname your_hostname`
Step 4	**ip domain-name** *domain_name* **Example:** `Switch(config)# ip domain-name your_domain`
Step 5	**crypto key generate rsa** **Example:** `Switch(config)# crypto key generate rsa`

- Set the ESXi host port (Fa0/15) to trunk, and the uplink to the SonicWall as a trunk

  Why: *Both need multiple VLANs over one cable. The 802.1Q tags tell the other side which VLAN the traffic belongs to.*

```
interface FastEthernet0/15
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

The switch only did layer 2. I left the routing to the SonicWall so any traffic between VLANs had to go through the firewall rules.

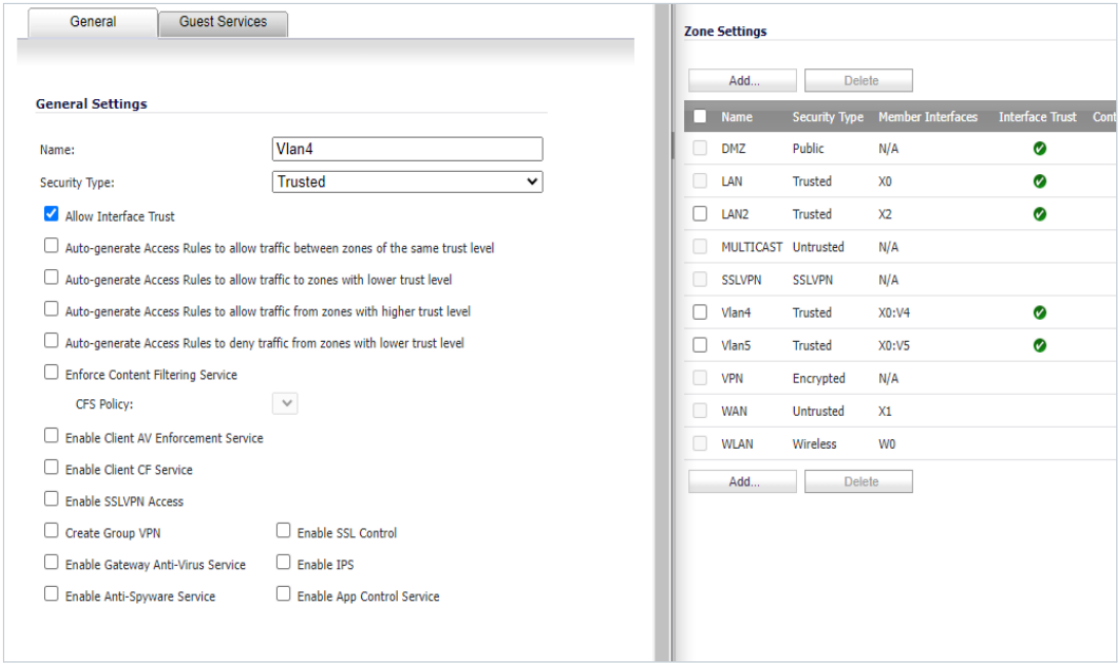
SonicWall

- Created VLAN sub interfaces on X0 (X0:V4, X0:V5)

  Why: *Router on a stick. One physical port carries multiple VLANs and each sub interface is the default gateway for its VLAN.*

- Zones: created a Trusted zone for each VLAN and did not check any of the auto generate rules. All access rules are set manually

  Why: *Rules are zone to zone. If the VLANs shared a zone they could talk by default. With the auto rules off everything is denied until I allow it, including internet.*



- Access rules: used the matrix to set access rules

  Why: *Only open what's actually needed between zones.*

- Inter-VLAN communication: used the matrix to create two way access rules

  Why: *The SonicWall is stateful, so replies to an allowed connection are allowed automatically. A rule in the other direction is only needed if that side has to start connections.*

- DHCP scope: Network > DHCP Scope > set a scope for the subnet before the DC was set up, then disabled it once the server was deployed

  Why: *Needed IPs before the server existed, and you don't want two DHCP servers handing out leases.*

VMware

- Installed ESXi 6.5 (ISO onto USB using Rufus) and set the root password during initial config
- Network config: set the management VLAN and a static IP for the host

  Why: *The host IP shouldn't change since vCenter and admins connect to it.*

- NIC teaming

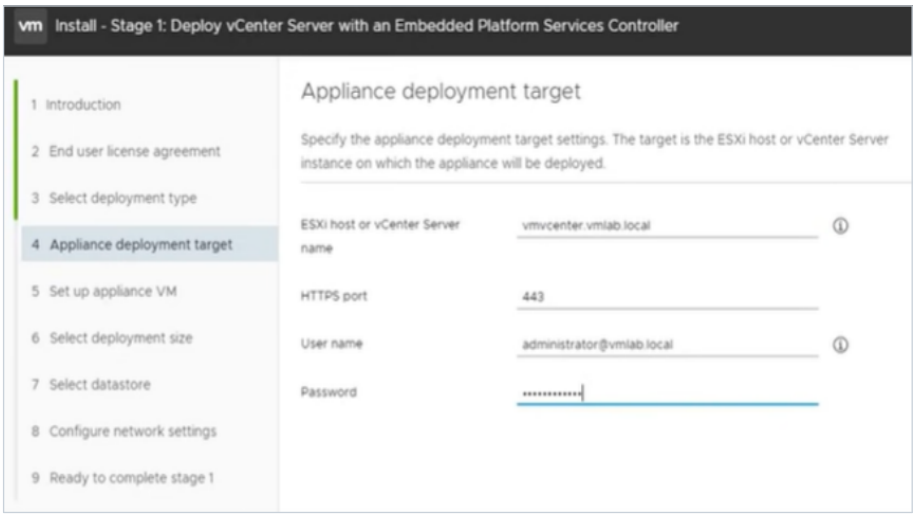
  Why: *If one NIC or cable fails the host stays connected.*

- NTP: Manage > Time > add NTP servers and start the NTP service

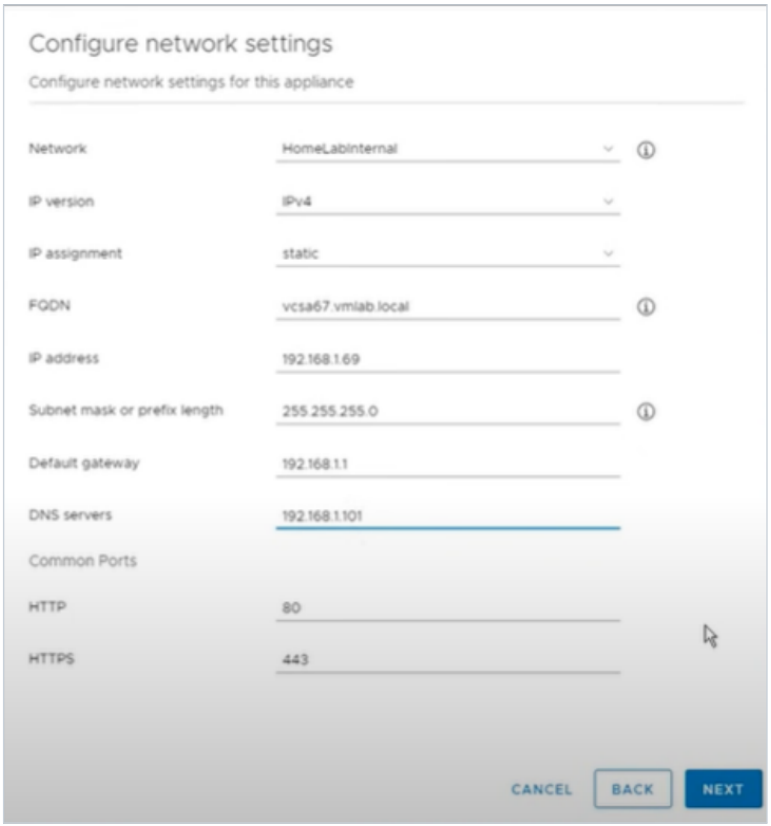
  Why: *AD (Kerberos) breaks if clocks are more than 5 minutes off, and logs from different machines are useless if the timestamps don't line up.*

- vSphere vCenter Appliance 6.7: mounted the ISO, ran the vcsa-ui-installer, chose Embedded Platform Services Controller, added the ESXi host as the deployment target

  Why: *Manage the hosts and VMs from one place.*



- Selected the compute resource, set the appliance VM name and password, deployment size tiny, selected the datastore and configured network settings



- Ran Stage 1, then Stage 2: time sync with the ESXi host, enabled SSH, created the SSO domain



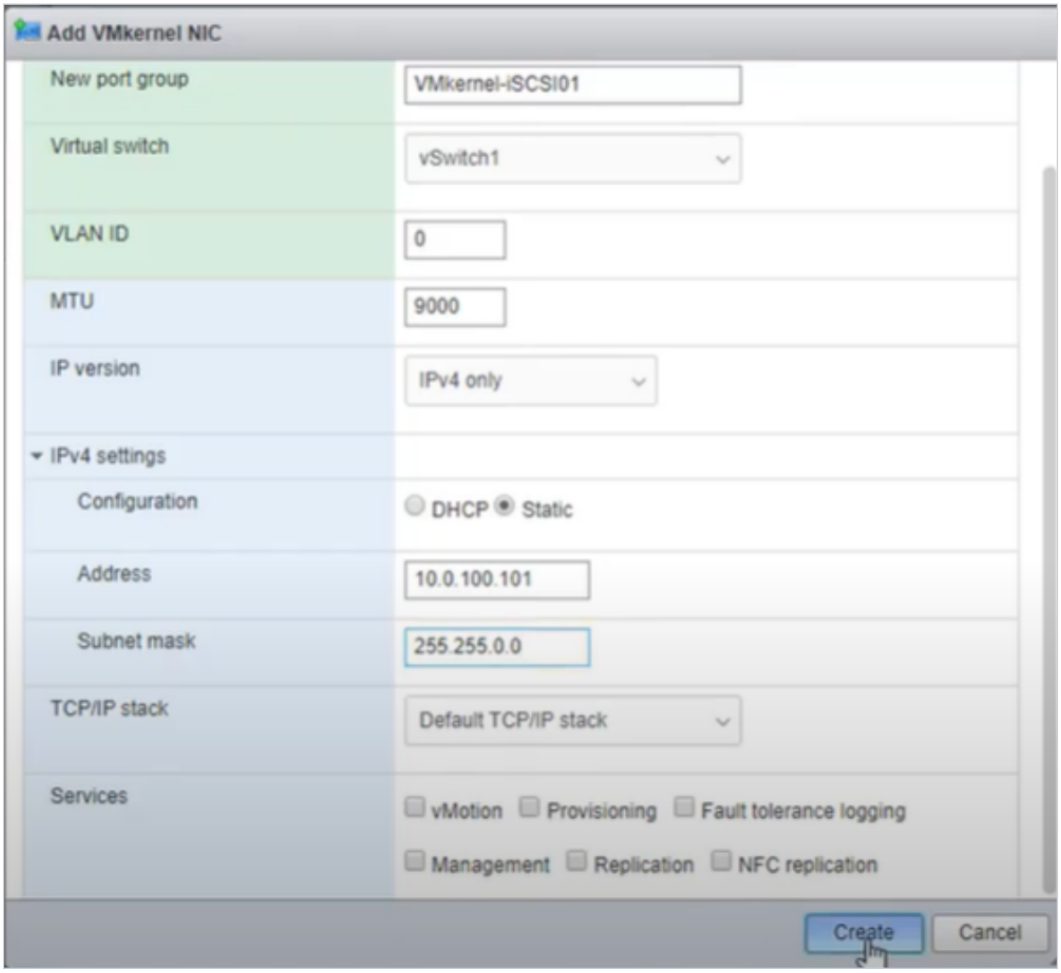
- Powered on the VM and added a PTR record for vCenter on the DNS server

  Why: *vCenter is picky about DNS. It needs both forward and reverse records or the install/login has issues.*

- Added a standard virtual switch: Networking > Add standard virtual switch

- Added a VMkernel NIC for iSCSI: new port group > assign to the virtual switch > VLAN > static IP, MTU 9000

  Why: *Keeps storage traffic separate from VM and management traffic. Jumbo frames (9000) cut overhead but have to be 9000 on every hop or packets drop. A VMkernel port is the host's own interface, VM traffic uses port groups.*



- Edited the port group > security > NIC teaming load balancing
- Configured the iSCSI adapter: Storage > Adapters > Software iSCSI > add dynamic target, then Storage > Devices > New datastore and formatted the disk

  Why: *ESXi connects to the NAS over iSCSI and the LUN gets formatted as a VMFS datastore to store the VMs on central storage.*

- Added the ESXi hosts to vCenter: new datacenter > add hosts

NAS

- Confirmed it picked up an IP, went to the portal and installed the TOS firmware
- Installed the iSCSI manager from the app store
- Created a target and LUN and set CHAP security

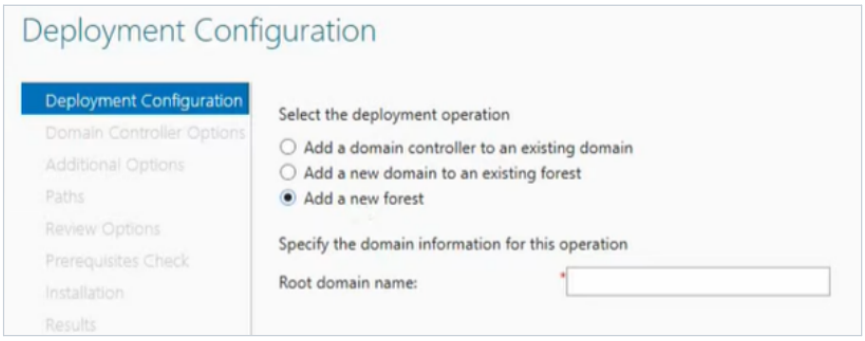
  Why: *The LUN is the chunk of disk ESXi uses. CHAP makes the host authenticate before it can connect to it.*

- Confirmed connectivity from the ESXi host

Windows Server

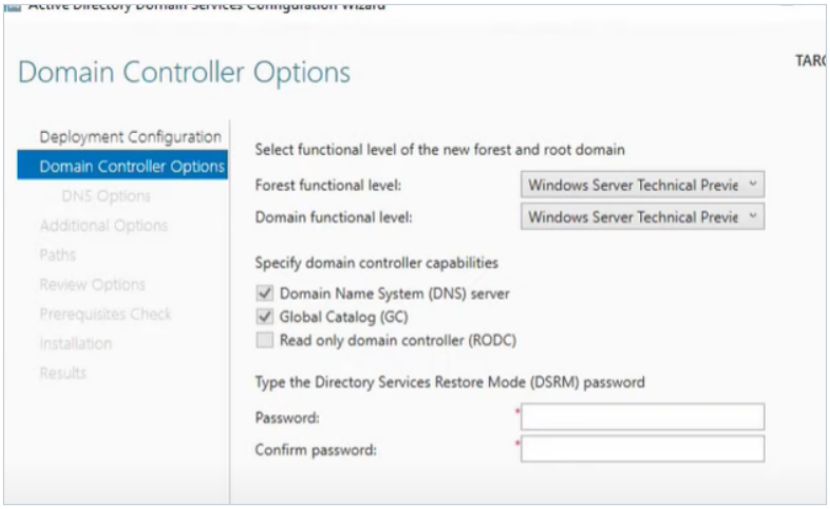
- AD: installed the AD role and set it up as a new forest

  Why: *First domain controller in the lab, so it creates a new forest and domain.*



- Set the DSRM password

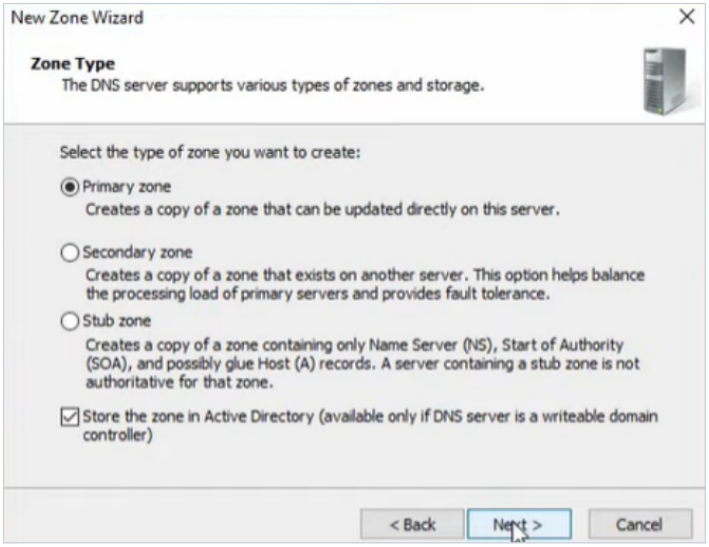
  Why: *DSRM is the recovery mode used if the AD database needs to be repaired.*



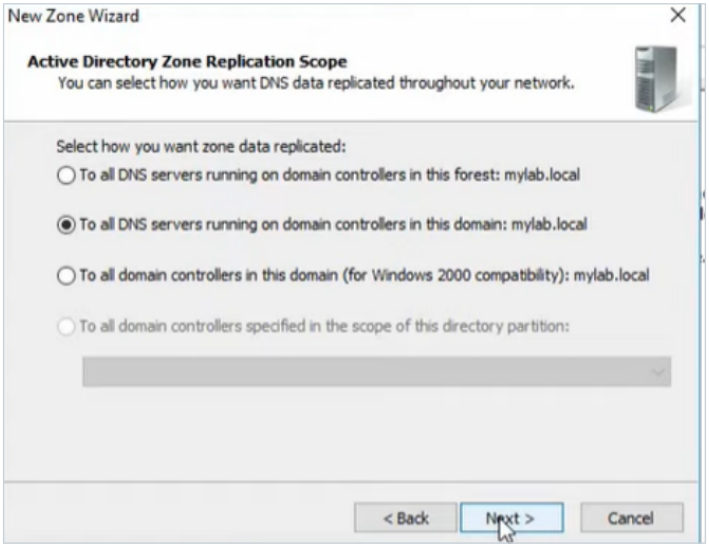
- Let the prereqs check out, installed and rebooted, then confirmed the server shows under Domain Controllers

- DNS: installed the role, opened DNS Manager and created a reverse lookup zone as a primary zone stored in AD

  Why: *AD relies on DNS to find the domain controller. Storing the zone in AD means it replicates to every DC.*

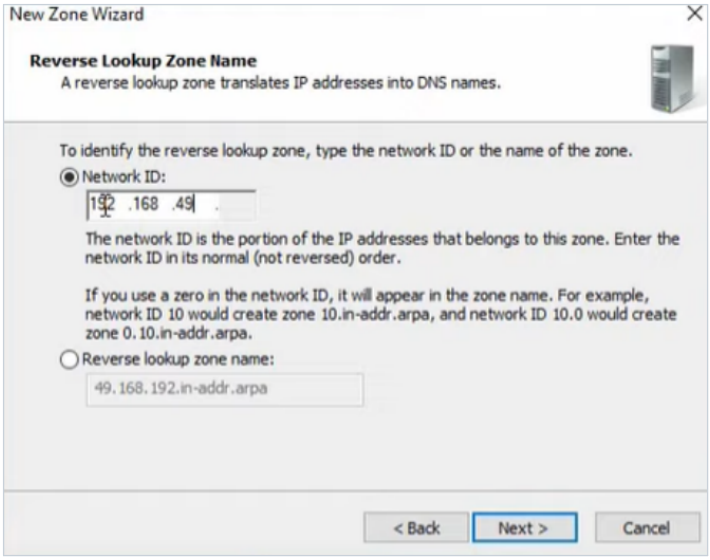


- Applied the zone to all DNS servers on the domain

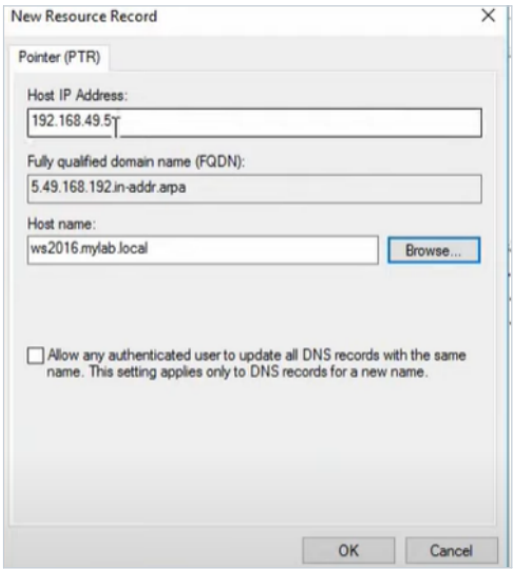


- Set the network ID to the subnet and went through the menu

  Why: *Reverse lookups resolve an IP back to a name.*



- Created a PTR record under the reverse lookup zone and added the server to the forward lookup zone

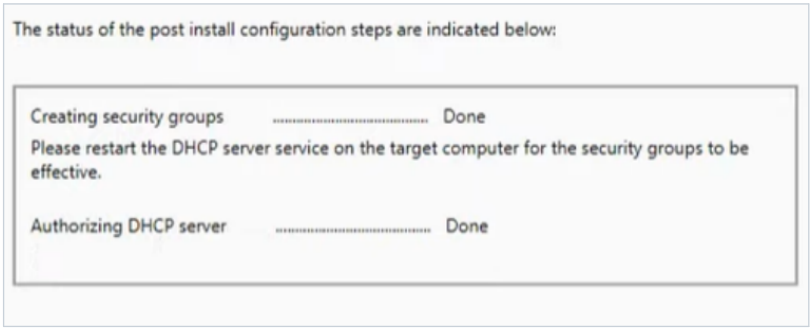


- Changed the server's preferred DNS to the DC's IP

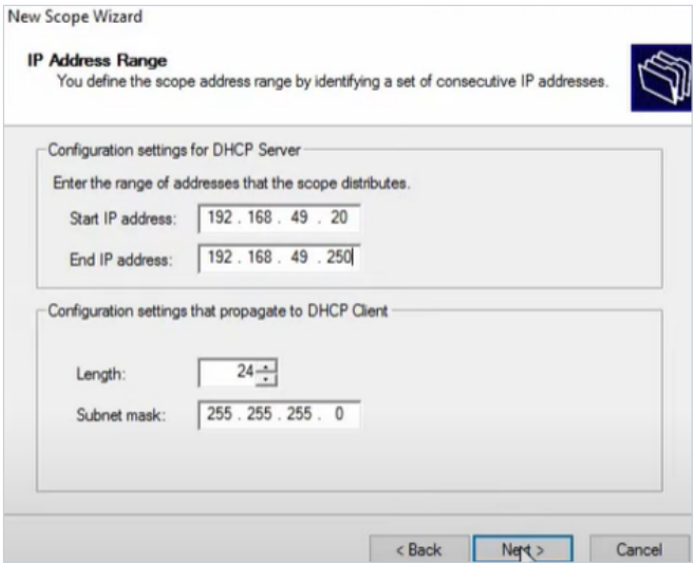
  Why: *Domain machines have to use the DC for DNS or domain joins and logons fail.*

- DHCP: installed the role, ran the post-install wizard and authenticated with domain admin creds

  Why: *Authorizing DHCP in AD stops rogue DHCP servers from handing out addresses.*

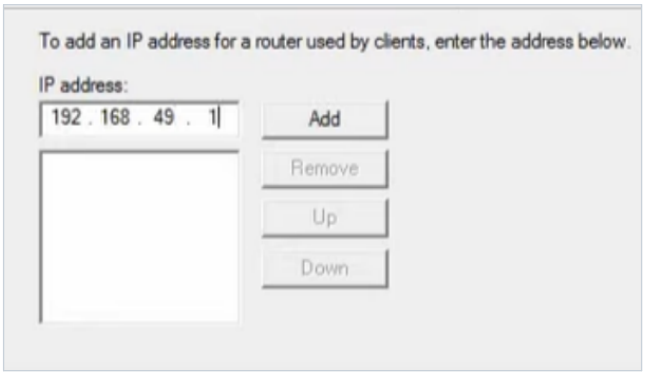


- Under IPv4 created a new scope, set the start/end range, length and subnet mask



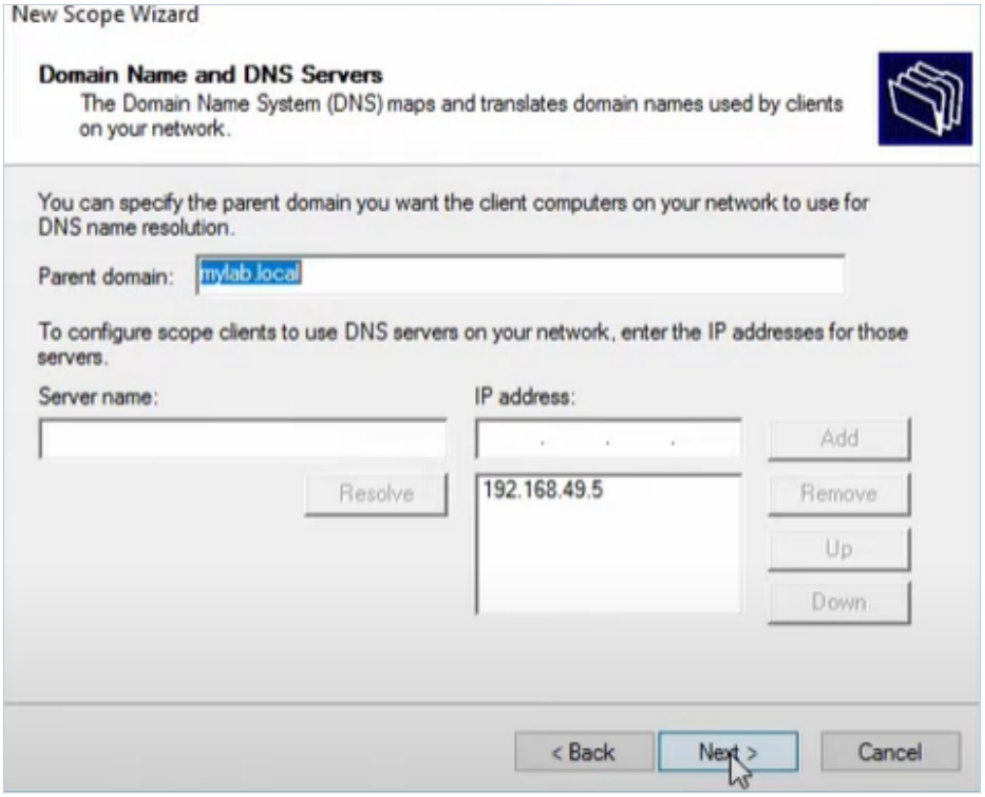
- Added exclusions and lease time and set the router (default gateway)

  Why: *Exclusions keep static devices out of the pool. The gateway is what clients use to leave their subnet.*



- Confirmed the domain name and DNS server, then activated the scope

  Why: *Clients get the DC as their DNS server so they can find the domain.*



Resources I used

AD DNS DHCP
https://www.youtube.com/watch?v=NE2nQlYcwao&t=891s

VMware ESXi iSCSI, vSwitch, VMkernel
https://www.youtube.com/watch?v=WvEU9WW6H9Y&t=652s

vCenter 6.7
https://www.youtube.com/watch?v=U-rilkWMkO4